Daily Review



1. Pressure equals force divided by…
2. F = 60 N, A = 300 mm^2. Find P.
3. The rod moves out. The exam word?
4. Before air on, Safety and the teacher check the…
5. The cylinder on a system diagram is the…

Area

0.2 N mm^{-2}

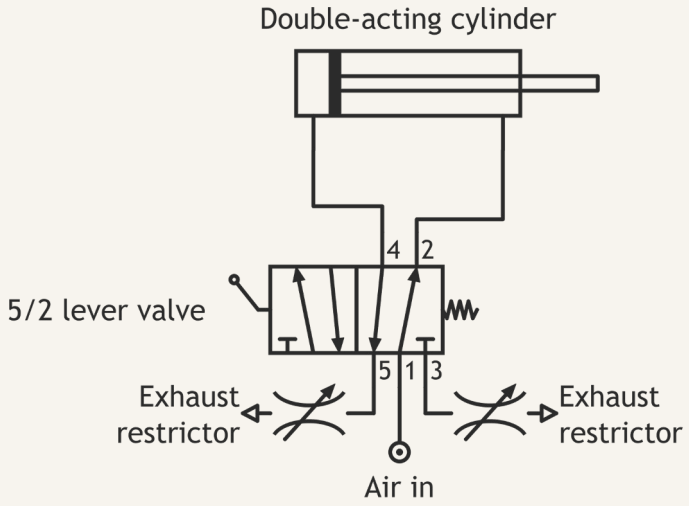
Outstroke

Air Permit

Output

Maintenance Night

Rides have broken overnight at Harbourlight.



S3 Engineering Science

The Job: Maintenance Night

i When the lever is pushed, the basket dips.

ii When the lever is let go, the basket lifts.

iii When a fault is fixed, it dips and lifts again.

Predict: tubes 4 and 2 swapped. Air on: what does the rod do?



By the end of this lesson…



1. I can find a fault by checking in order
2. I can write a fault, its cause and its fix
3. I can hide a fault safely, with the air off

THE BUILD

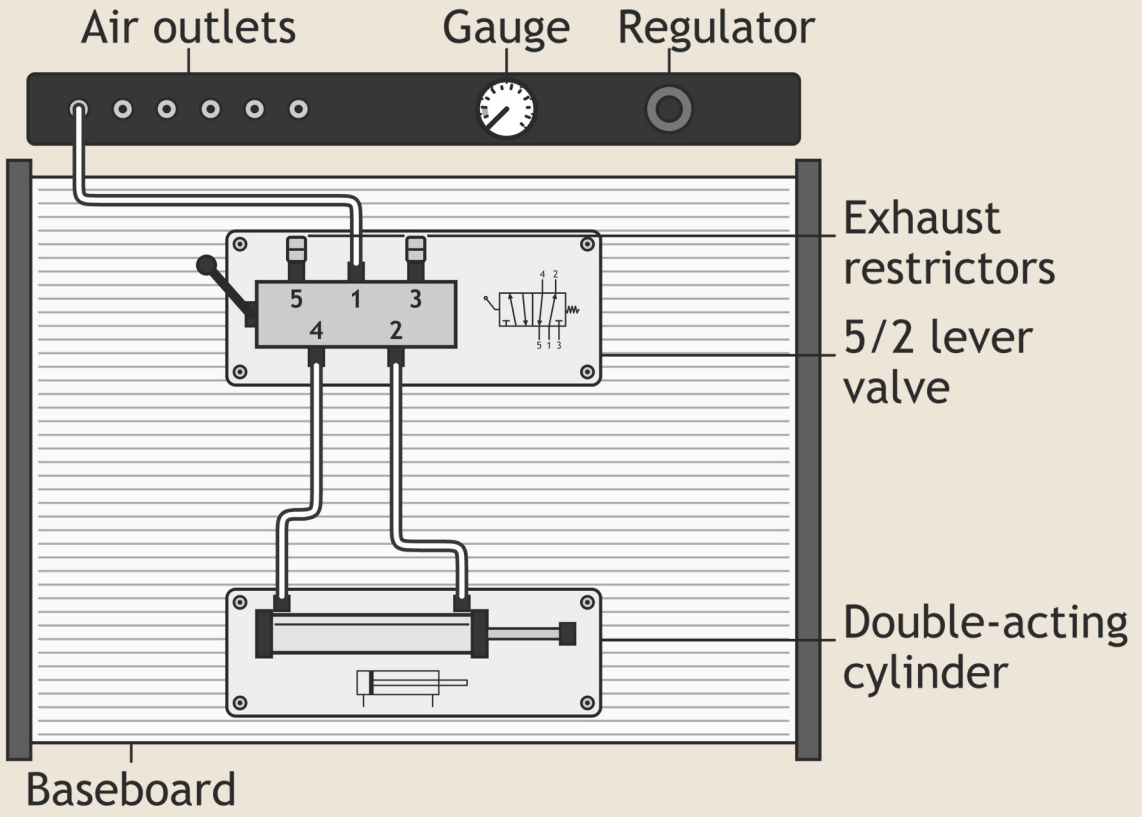
Find and Fix

Build, hide, swap, find, write it up.

What to Collect

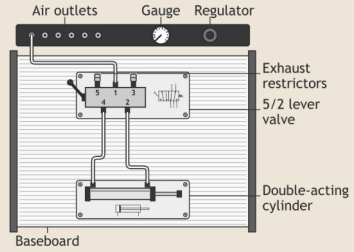
Builder and Safety: one set per team.

1 Bench card 4
2 Manifold (air supply)
3 5/2 lever valve
4 Double-acting cylinder
5 2 exhaust restrictors
6 Tubing
7 Goggles, one each



Build the Paint Dip

Task 1 (B6A-L6-p18)



Build from bench card 4.

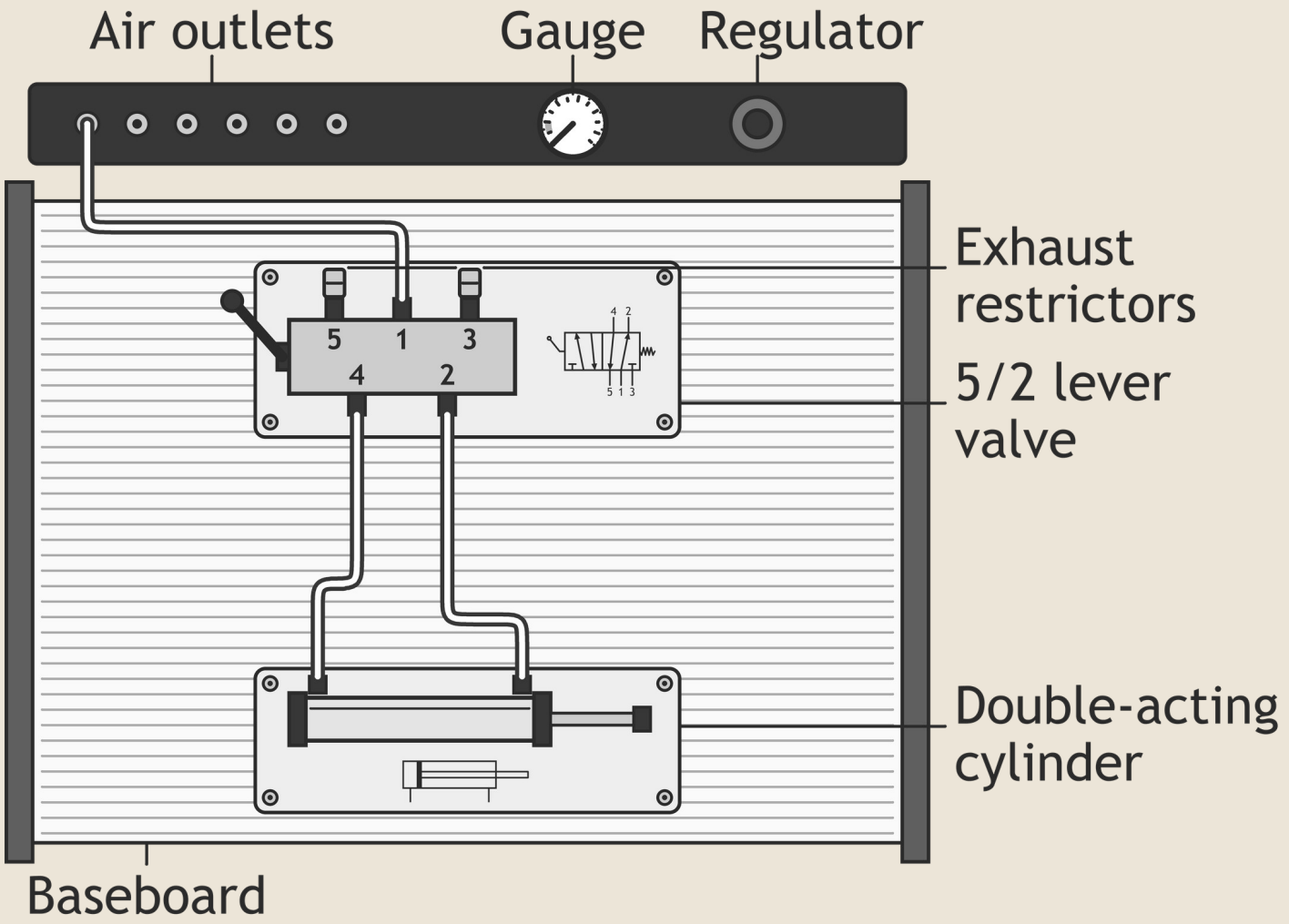
Air Permit, then test points i and ii.

Regulator back to 0.

SAFETY Regulator at 0 to build. 1 to 2 bar to test.

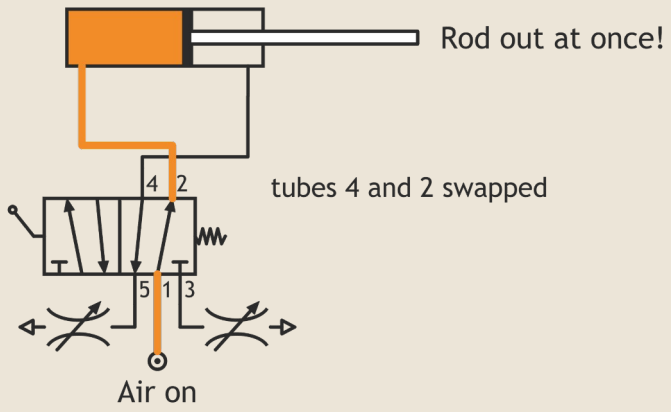
10 min

Build It: Bench Card 4



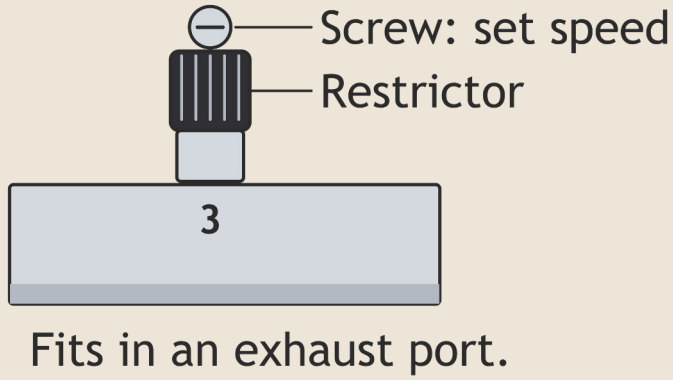
The Only Faults You May Hide

Regulator at 0. One fault. Never damage a part.



Swap 4 and 2

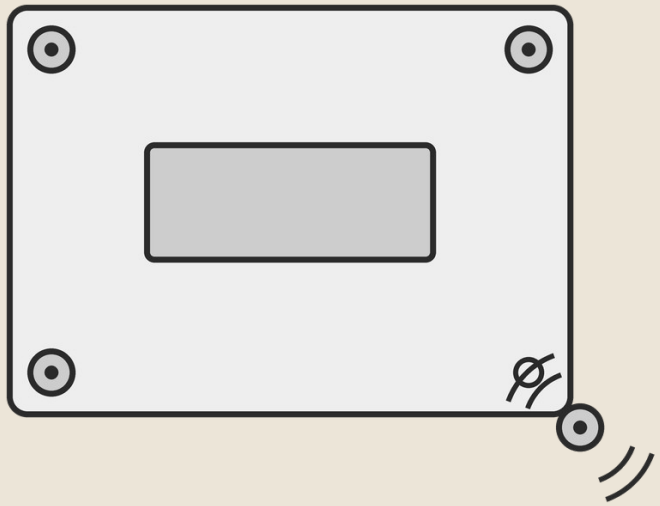
a tube



Fits in an exhaust port.

Restrictor in

a setting

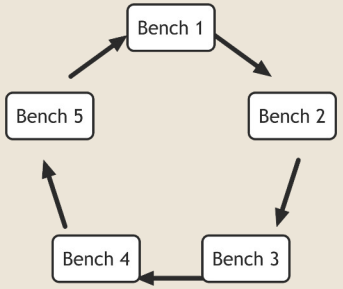


Loose bolt

a part

Hide One Fault

Task 2 (B6A-L6-p19)



Regulator at 0.
Hide ONE fault from the list. Tick it.
Close your booklet. Move on one bench.

SAFETY Never damage a part. Nothing off the list.

4 min

Check in Order

Check: yes, go on down	If no: fix it
1 Regulator at 0? Hands clear?	Stop. Do this first.
2 Every tube pushed in fully?	Push it in.
3 Every tube on the right port?	Move it.
4 Both restrictors open a little?	Open it a little.
5 Every part bolted down tight?	Tighten the bolt.
6 Teacher checks. Does it work?	Air off. Back to 1.

Show Me: Which Check?

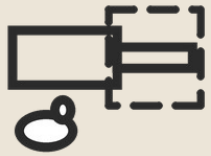
1	Before you touch anything.	Check 1: regulator at 0
2	A tube pops out with a hiss.	Check 2: pushed in
3	The rod goes out the moment the air is on.	Check 3: right port
4	The rod crawls out very slowly.	Check 4: restrictors

Fault-Find Permit

No permit, no air. Initials before you look.



Air off
while we look



Hands clear
of the piston rod



Teacher checks
before air on

Find the Fault

Task 3 (B6A-L6-p19)

Fault-find permit: initials first.
Check in order, 1 to 6.
Teacher checks before air on.

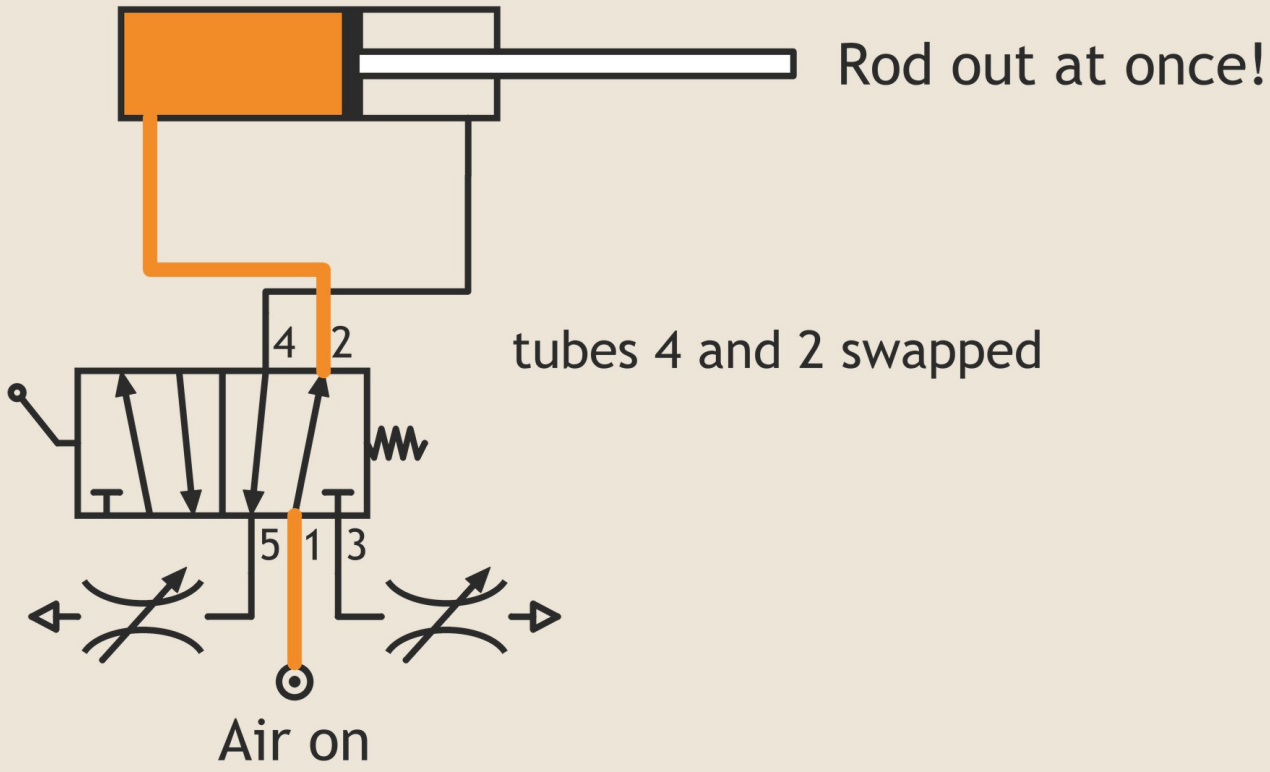
SAFETY Air off while you look. Hands clear of the rod.

8 min

Test It: What Did You Find?

1	Your dip before the swap: points i and ii met?	Yes: it dipped and lifted
2	Tubes 4 and 2 swapped. What did the rod do?	Shot out at air on
3	A restrictor right in. What did the rod do?	Very slow, or stuck
4	Point iii met? Which check found it?	Yes: check 3, 4 or 5

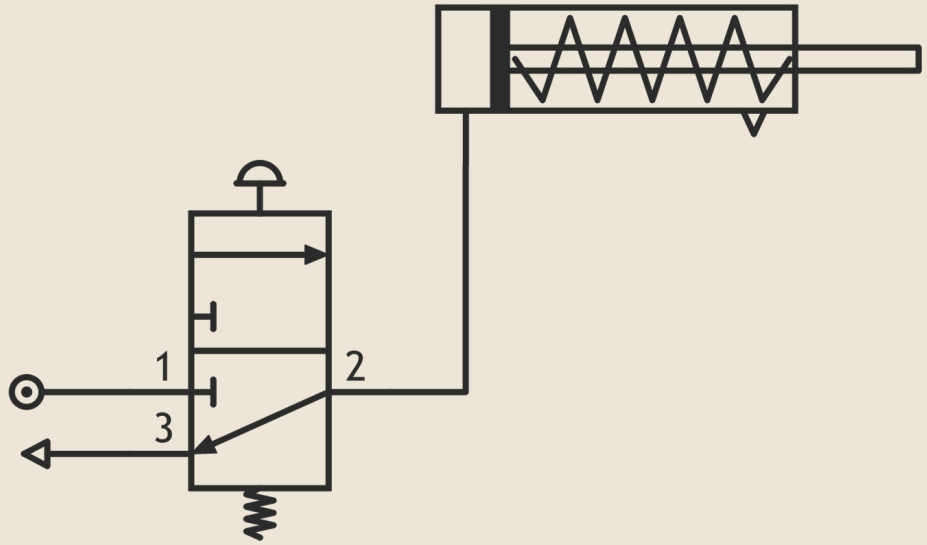
Why the Air Must Be Off



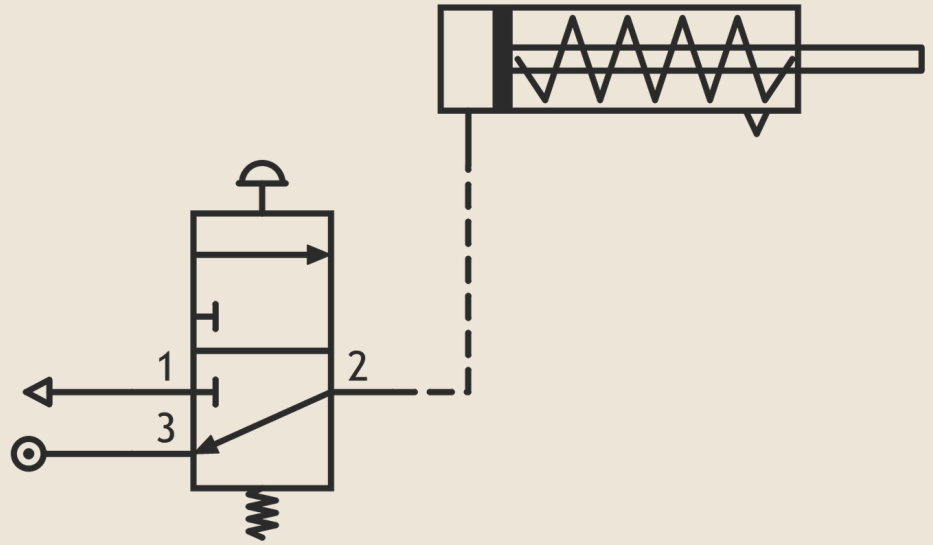
1 Right tubes, air on: the front fills. The rod stays in.

2 Tubes 4 and 2 swapped, air on: the back fills. The rod shoots out at once.

Spot the Mistakes



Right



Two mistakes: what?

Air in and exhaust are on the wrong ports

The main pipe is dashed: it should be solid

Write It Up and Questions

Tasks 4-6 (B6A-L6-p20)

Task 4: fault, cause, fix.
Task 5: Questions 1 and 2 on your own.
Task 6: Key words.
Question 3 waits for Lesson 8.

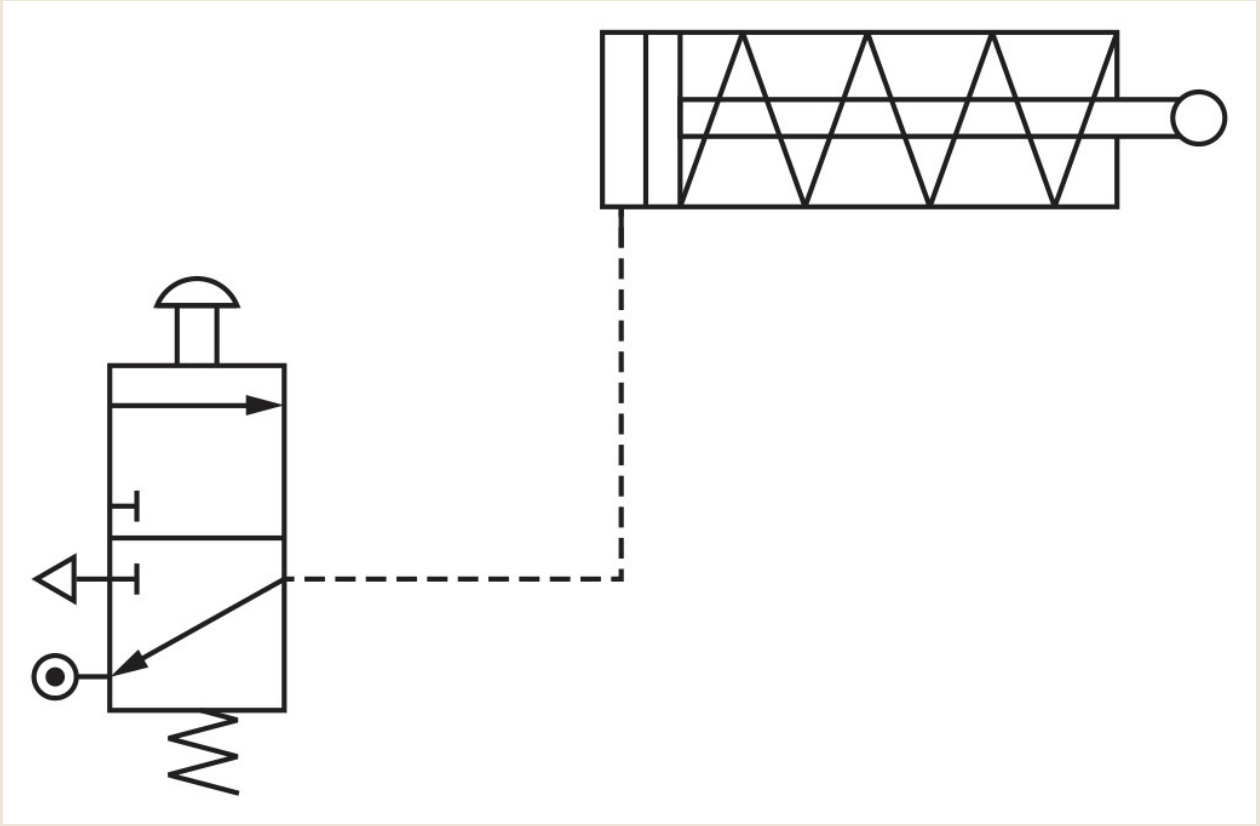


8 min

Answers: Tasks 5-6 (B6A-L6-p20)

Question	Mark 1	Mark 2
Q1 (2 marks)	Parts move suddenly	Could trap fingers
Q2 (2 marks)	Rod goes out at once	Lever: it comes back
Key words	1 0, 2 port	3 cause

Exam Practice



Task 7 (B6A-L6-p21): exam question, 2 marks.
Describe TWO errors in the diagram.

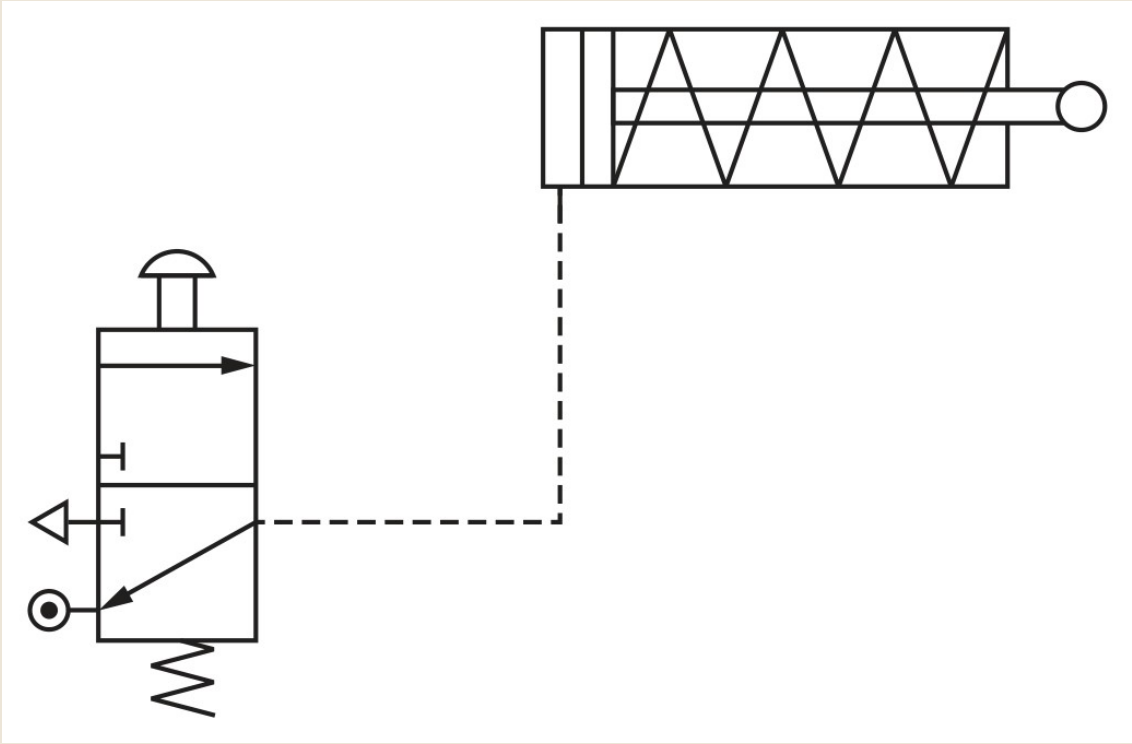
4 min

Answers: Task 7 (B6A-L6-p21)

Describe two errors that were made in this circuit diagram.

Air in and exhaust are on the wrong ports (1)

The line to the cylinder should be solid (1)



Plenary: Maintenance Night



1	Look for a fault: the regulator at…	0
2	Check 1, every time?	Regulator at 0
3	Tubes 4 and 2 swapped: the rod…	Goes out at once
4	Write up three things. Name them.	Fault, cause, fix
5	A main pipe on a diagram is drawn…	Solid

Look back — by the end of this lesson…



1. I can find a fault by checking in order
2. I can write a fault, its cause and its fix
3. I can hide a fault safely, with the air off

LOOKING AHEAD

Lesson 7: The Train Door

Two ways in, one brake: can air decide?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Paint dip circuit: The paint dip circuit from the Matrix CP2080 Pneumatics Essentials Worksheet 4; the checks adapted from Worksheet 8 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Paint dip circuit, kit and checks: The paint dip circuit from the Matrix CP2080 Pneumatics Essentials Worksheet 4; the checks adapted from Worksheet 8 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Exam Practice and Model Answer: 2024 Q3(a): © Qualifications Scotland (SQA). N5 Engineering Science 2024, Question 3(a) and its marking instructions (booklet Task 7).

R Stewart - October 2026